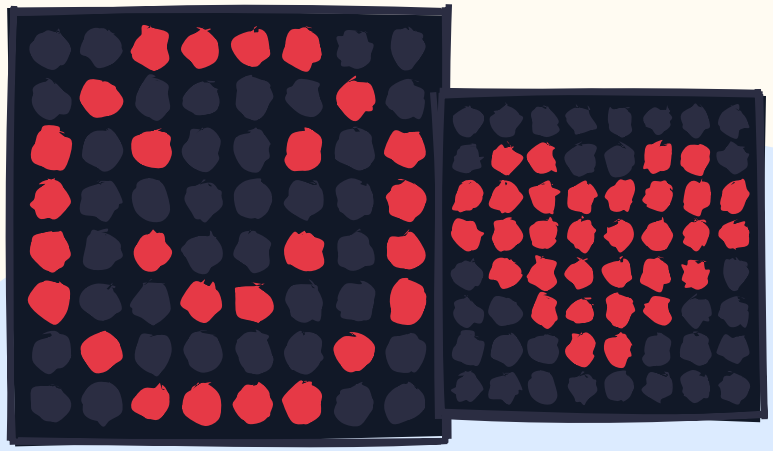
Junior Makers · Class 22

8×8 LED Matrix
Emoji Display

64 lights, endless pictures

Rows & columns | Binary pictures | Animation



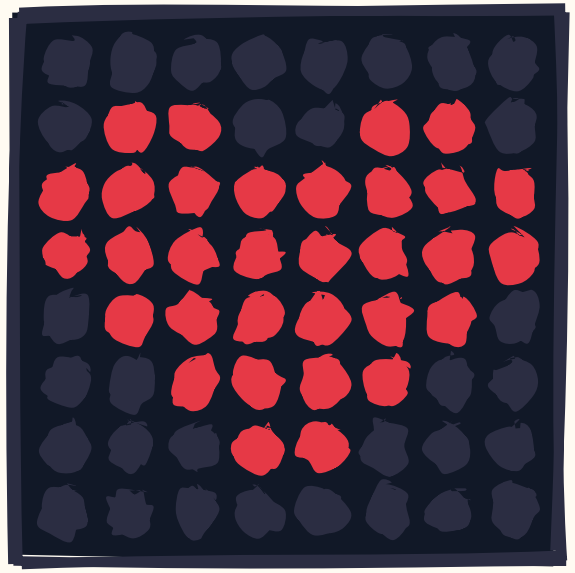
Did you know?
The scoreboard at a cricket stadium is a giant grid of LEDs, controlled the same way as our little 8×8 matrix.

Let's think...

How do bus signs show words?

Dots everywhere

Bus signs, shop signs and stadium scoreboards show words, numbers and even pictures. Look closely: it's all **dots** that are either on or off. How do you tell 64 dots what picture to make?



One wire each?

Draw with a pen?

Rows of 0s and 1s!

Today: you design **emojis** and **animations** for an 8×8 LED matrix and scroll your own message.

Learning objectives

By the end of this class I can...

1	**Rows & columns:** find any LED on an 8×8 grid by its **row and column**	**Proof:** I can light row 2, column 5 when asked
2	**Pictures as bits:** turn a drawing into **8 rows of 0s and 1s**	**Proof:** my emoji drawing has 8 binary rows
3	**Animation:** make an **animation** by showing frames one after another	**Proof:** my emoji blinks or moves
4	**Message board:** **scroll** a message and design a useful sign	**Proof:** my matrix scrolls a message for our class

Objective 1 · Rows & columns · video

Video time!



How Pixels Work

In One Lesson · about 3 minutes · youtu.be/0B79dGR19Tg

Watch for...

1. What is a **pixel**?
2. Why do pictures look smooth when there are lots of pixels?
3. How does a screen make so many colours?

Scan to watch

After the video: How many pixels does our matrix have? Would a photo of your face be recognisable on it?

Objective 1 · Rows & columns

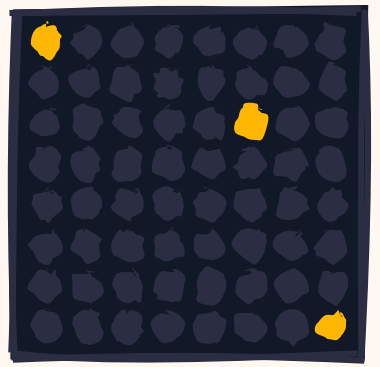
Rows and columns, starting at 0

The idea

The matrix has **8 rows** (across) and **8 columns** (down): 64 LEDs. Each LED has an address: **(row, column)**, numbered **0 to 7**.

Row 0 is the top. Column 0 is the left. So (0, 0) is the top-left corner and (7, 7) is the bottom-right.

Example: seats in a cinema



Your cinema ticket says 'Row F, Seat 12'. With just two pieces of information you find your exact seat among hundreds. Every LED has a 'ticket' too: its row and column.

Objective 2 · Pictures as bits

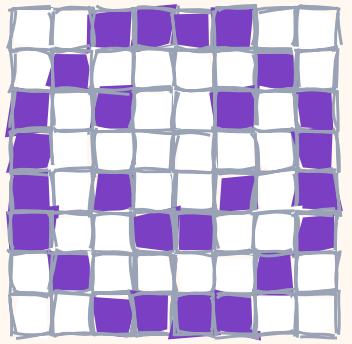
A picture is 8 rows of bits

The idea

Each row is 8 LEDs: **1 = on**, **0 = off**. So one row is 8 bits: **one byte** (remember JM-19's bracelets?).

A whole smiley is just **8 bytes**:
00111100 · 01000010 · 10100101 · 10000001 · 10100101 · 10011001 · 01000010 · 00111100

Example: cross-stitch patterns



00111100
01000010
10100101
10000001
10100101
10011001
01000010
00111100

Embroidery patterns are grids: each square is either stitched or not. Your grandmother could copy a design just from a grid of filled and empty squares. That's exactly our emoji grid.

Objective 3 · Animation

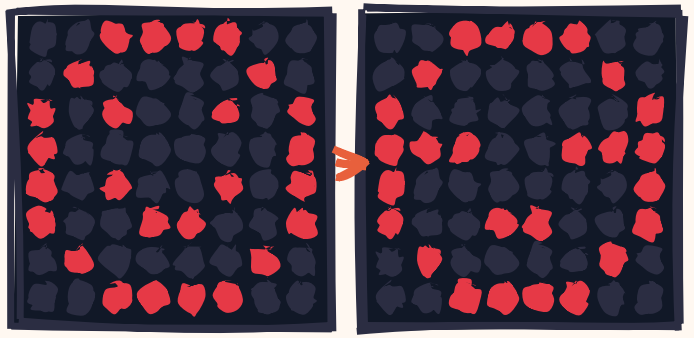
Animation = frames shown fast

The idea

A **frame** is one picture. Show frame 1, wait, show frame 2, wait… Your brain joins them into **movement**.

A **blink** needs just 2 frames: eyes open, eyes closed. A **bouncing ball** might need 6. Short waits (100–200 ms) look smooth; long waits look jumpy.

Example: a flipbook



Draw a stick figure in the corner of every page of a notebook, each a tiny bit different. Flick the pages: it moves! Cartoons and games work the same way, with 24 to 60 frames every second.

Objective 4 · Message board

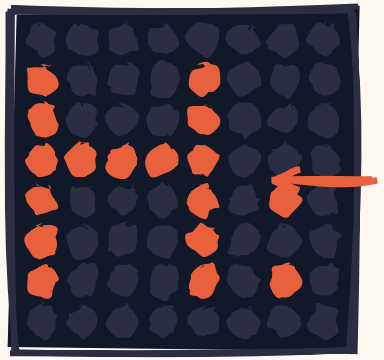
Scrolling messages

The idea

Words are too wide for 8 columns, so we **scroll**: shift every column one step left, add a new column on the right, repeat. ArduinoBlocks has a **scroll text** block that does this for you.

Choose the **speed** carefully: too fast and nobody can read it, too slow and people walk past.

Example: a news ticker



TV news channels show a moving line of headlines at the bottom of the screen. You can read a long sentence in a small space because it slides past you.

Real world

LED matrices all around



Scoreboards
Stadium screens are huge LED grids



Bus and train signs
Show the next stop and arrival time



Shop signs
'OPEN' signs and moving price displays



Clocks
Digital clocks and timers in airports



Design a sign for our school: what message would help people most?

Pixels
Your phone screen is the same idea, with millions of tiny RGB dots instead of 64 red ones.

Quick quiz

Bits and pictures

Question	Answer
1. How many LEDs on an 8×8 matrix?	
2. One row of 8 LEDs is how many bits?	
3. Write a row with only the two middle LEDs on.	
4. What address is the bottom-left LED?	
5. How many bytes for one full picture?	

Answers

Cover! 1: 64 · 2: 8 (one byte) · 3: 00011000 · 4: (7, 0) · 5: 8 bytes

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Human matrix
2. Design your emoji on paper
3. Wire the matrix
4. Show your emoji
5. Make it blink
6. Make a useful sign
7. Bug hunt

Class game

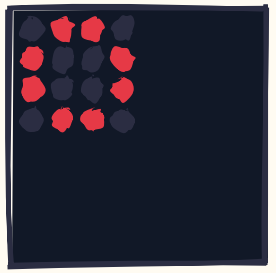
Human matrix

How to play

1. Make a 4 × 4 grid of 16 students sitting on chairs.
2. Each holds a card: **bright side = ON**, dark side = OFF.
3. The 'computer' reads out rows as 0s and 1s: **Row 0: 0110!**
4. Students in that row show ON or OFF to match.
5. After 4 rows, what picture did you make?

Then discuss

How many numbers did the computer need to say to make a whole picture?



Big idea

A picture can be sent as rows of 0s and 1s. That's how matrices, screens and printers work.

Objective 2 · Build step 1 of 4

Design your emoji on paper

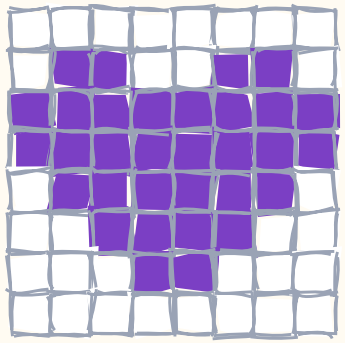
a Take the 8×8 template. Shade squares to draw an emoji or icon.

b Keep it bold: thin details disappear on 64 dots.

c Write each row as 8 digits: shaded = 1, empty = 0.

d Check: every row must have exactly **8** digits.

e Swap with a partner: can they draw your emoji from just the numbers?



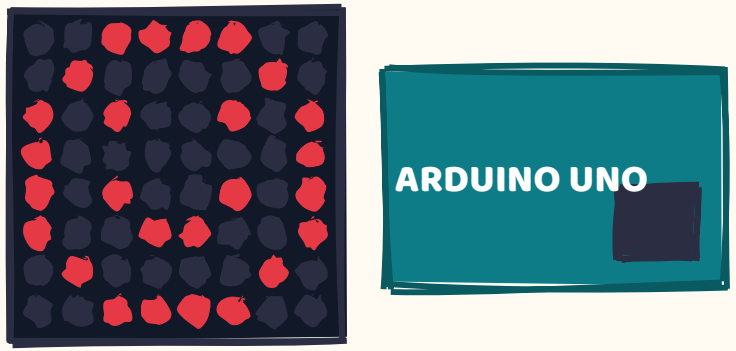
```
00000000
01100110
11111111
11111111
01111110
00111100
00011000
00000000
```

Check: Your partner redraws your emoji perfectly from the 8 rows of numbers.

Objective 1 · Rows & columns · Build step 2 of 4

Wire the matrix

Matrix pin	Arduino
VCC	5V
GND	GND
DIN	pin 12
CS	pin 10
CLK	pin 11



The MAX7219 chip

Controlling 64 LEDs would need 64 wires! The chip on the back does the work: we send it just 3 wires of data.

Objective 2 · Pictures as bits · Build step 3 of 4

Show your emoji

```
Setup
    LED matrix init DIN 12 CLK 11 CS 10
    LED matrix set intensity 4
Loop
    LED matrix draw image [8×8 grid editor: your emoji]
    wait 2 seconds
    LED matrix clear
    wait 0.5 seconds
```

Grid editor

In ArduinoBlocks, click the squares in the image block to copy your paper design. Or type your 8 binary rows.

Upside down?
Modules are mounted differently. If your emoji is rotated, turn the matrix or flip the design.

Objective 3 · Animation · Build step 4 of 4

Make it blink

- Loop
  - LED matrix draw image [face, eyes OPEN]
  - wait 2 seconds
  - LED matrix draw image [face, eyes CLOSED]
  - wait 150 milliseconds
- (bonus) repeat 8 times: scroll image left by 1

Timing	Looks like...
Long wait open, short closed	
Equal waits	
Very short (20 ms)	

Real blinks

People blink for about 0.1 to 0.4 seconds. That's why 150 ms looks natural!

Objective 4 · Message board · design

Make a useful sign

Sign	Shows	Animation
Welcome sign	scroll 'WELCOME TO ROBOTICS'	+ smiley
Quiet please	a 'shh' face	blinks slowly
Timer	countdown 5, 4, 3, 2, 1	then a star
Your idea		

Buzzer bonus

Beep the buzzer when the message starts, so people look!

Test like an engineer

Bug hunt

Bug	Likely cause	Did it happen?
Nothing lights	VCC/GND swapped or DIN/CLK/CS wrong pins	
All 64 LEDs on	matrix not initialised / wrong CS pin	
Emoji upside down	module rotated	
Very dim or very bright	intensity setting (0–15)	
Scroll unreadable	speed too fast	

Debug checklist

Light just ONE LED at (0, 0) first. Then you know where the top-left corner really is.

Change one block at a time and test again.

Level up!

Matrix challenges

Choose your challenge



Heartbeat: a heart that grows and shrinks (3 frames).



Pac-Man: an animated mouth that opens and closes as it moves.



Countdown: show 9 down to 0, then a smiley and a beep.



Rain: dots fall from the top row to the bottom, at random columns.

Think like an engineer

How many bytes do you need to store a 10-frame animation on an 8×8 matrix?

Boss level

Join two teams' matrices side by side and scroll one long message across both!

Objective check

Can I really do it? Prove it.

01	Point to LED (2, 5) on the matrix. How did you find it?	○ ○ ○
02	Read out your emoji's 8 rows of bits.	○ ○ ○
03	Show your animation. How many frames, and what waits did you use?	○ ○ ○
04	Show your scrolling sign. Who would it help?	○ ○ ○

Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Emoji guessing game: show your emoji for 2 seconds. Can the class guess it?

Show what you know

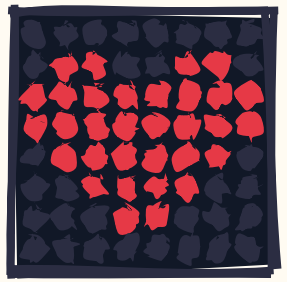
What did we learn?

- [x] Every LED has an address: (row, column), from 0 to 7
- [x] Each row is 8 bits: one byte; a picture is 8 bytes
- [x] Animation = frames shown one after another
- [x] Scrolling shifts columns to show long messages
- [x] The MAX7219 chip controls 64 LEDs with 3 wires

Exit ticket: write the 8 bits for a row where only the first and last LEDs are on.

At home

Find an LED sign or scoreboard near you. Can you see the dots? Count how many rows it has.



Next class:
Ultrasonic Theremin: Music in the Air!